DevOps Lab Program

Program8:Create kubernetes Pods using Yaml Descriptors

Structure of Program

Program8/

→nodejs-configmap.yaml

→nodejs-pod.yaml

→server.js

Steps for execution:

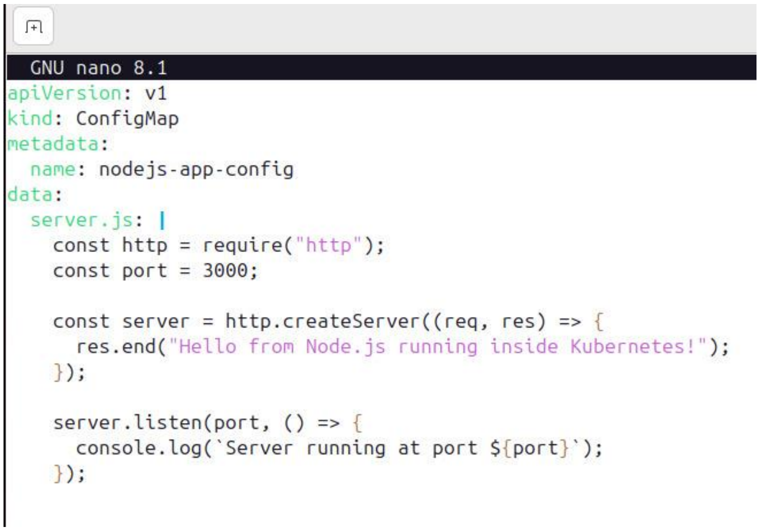
Step 1: Create a server.js file

```
GNU nano 8.1
const http = require("http");
  const port = 3000;

  const server = http.createServer((req, res) => {
    res.end("Hello from Node.js running inside Kubernetes!");
  });

  server.listen(port, () => {
    console.log(`Server running at port ${port}`);
  });
```

Step 2: Build the config map file

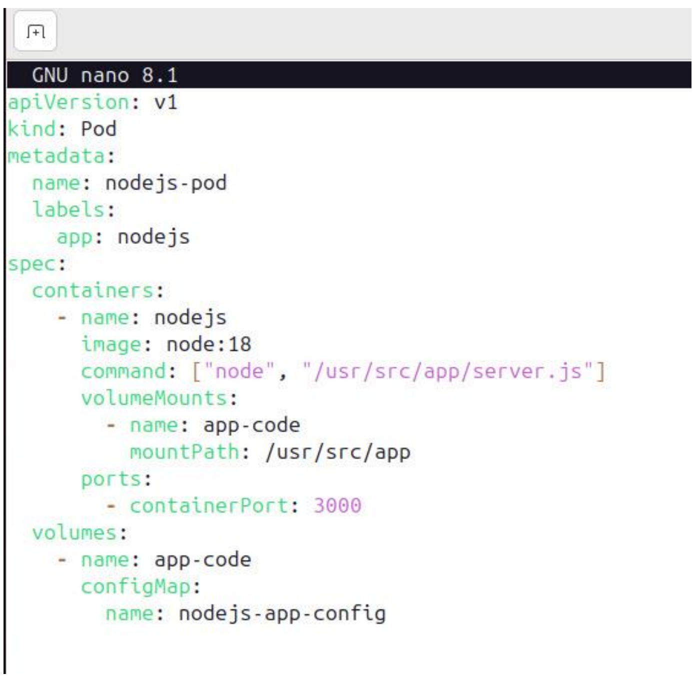


```
GNU nano 8.1
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Build the pod.yaml file



```
GNU nano 8.1
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4: Apply yaml files

Kubectl apply –f nodejs-configmap.yaml

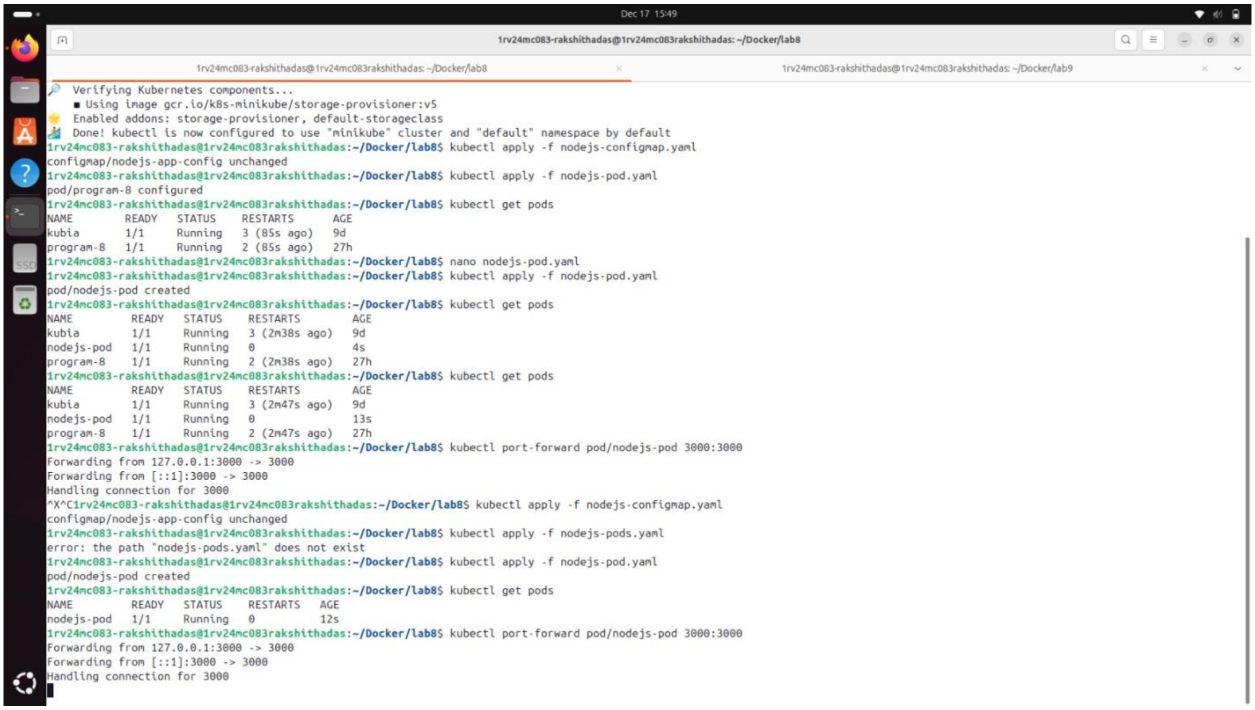
Kubectl apply –f nodejs-pod.yaml

Step 5: Check status

Kubectl get pods

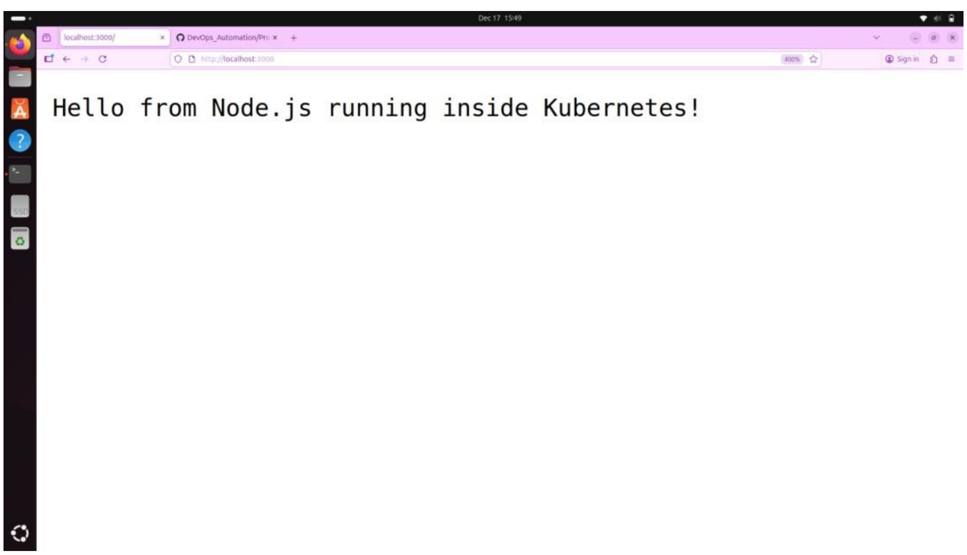
Kubectl get deployments

Step 6: Access the web page



Step 7: Check the output in:

http://localhost:3000



Step 8: Delete all the pods

Step 9: Delete all the services

Step 10: Delete all the deployments